

Question 1:(7 points) Given the following code Answer the following

```
class TypeA{
private:int x,*p; char c;
public:
    TypeA(int a) { x = a; }
    TypeA(int *q){ p = q; }
    TypeA(char c){ c = c; }
    void fun1(int a){ cout<<x+a; }
    int * fun2(int *k){
        *k+=*p;
        return k; }
    char & fun3(char m){
        char *g=new char(m);
        return *g; }
    int fun4(TypeA k){
        return pow(*(k.p),2); }
    char & fun4(char &h){
        if(h=='a') return ++h;
        else return --h; }
};
```

1- output if these instructions were executed in main

```
TypeA xa(1),xb(new int(7)), xc('c');
int x=2; int *p=&x;
p=xb.fun2(new int(2)); char cc='b';
cout<<xa.fun4(cc);
```

A. a B. b C. cc D. other

2- How many overloaded constructor in the class TypeA?

A. 4 B. 5 C. none D. other

3- which function returns a reference

A. fun1 B. fun2 C. fun3 D. other

4- Which function returns a pointer?

A. fun1 B. fun2 C. fun3 D. other

5- output if these instructions were executed in main

```
TypeA xa(1),xb(new int(7)), xc('c');
cout<<xa.fun4(xb);
```

A. 16 B. 49 C. cc D. other

6- Output if these instructions were executed in main?

```
TypeA xa(1),xb(new int(7)), xc('c');
int x=2; int *p=&x;
p=xb.fun2(new int(2));
cout<<xc.fun3('d');
```

A. c B. d C. e D. other

7- what is the output if these instructions were executed in main

```
TypeA xa(1),xb(new int(7)), xc('c');
int x=2; int *p=&x;
p=xb.fun2(new int(2));
cout<<*p;
```

A. 9 B. d C. 7 D. Other

Question 2: (5 points)

Given the following code answer the questions

```
template <class T, class U >
int fun(T a, U b){ return a+b;}

class A{
private: int x=5,y=2;
public:
A(){ cout<<x+y; }
A(int a, int b):x(a),y(b){}
void set(int a,int b){ x=fun(a,b);}
void get(){
int x=3;
cout<<this->x + x<<" "<<this->y; }
A * fun1(A *m){
this->x= m->x;
this->y= m->y;
return this; }
void out(){
cout<<this->x + this->y; }
~A(){};
```

1- What is the output if these instructions were executed in main?

```
int main(){
int x=1,y=2; cout<<fun(x, y);
return 0;}
```

Answer 3

2- What is the output if these instructions were executed in main?

```
int main(){
int x=1,y=2;
A obj(x,y); obj.set(3,4);
obj.get();
return 0;}
```

Answer 10,2

3- What is the output if these instructions were executed in main?

```
int main(){
int x=1,y=2;
A obj(x,y);
obj.fun1(new A(4,5))->out();
return 0;}
```

Answer 4

4- What is the output if these instructions were executed in main?

```
int main(){
A();
A xa;
xa.out();
return 0;}
```

Answer 777

5- How many times the destructor will be called when these instructions are executed?

```
int main(){
A obj(1,2), xa, *xb=new A();
```

Answer 3

```
return 0;}
```

Question 3: (3 points)

What will happen in the following C++ code snippet?

```
1-
int a = 100, b = 200;
int *p = &a, *q = &b;
p = q;
```

A. b is assigned to a

☒ B. p now points to b

C. a is assigned to b

D. q now points to a

```
2-
int main() {
    int a = 5, b = 10, c = 15;
    int *arr[] = {&a, &b, &c};
    cout << arr[1];}
```

A. 5

B. 10

C. 15

☒ D. it will return some random number

```
3-
int main() {
    char *ptr;
    char str[] = "abcdefg";
    ptr = str;
    ptr += 5;
    cout << ptr;
    return 0;}
```

☒ A. fg

B. cdef

C. defg

D. abcd

Question 4: (5 points)

Write a class so that the following instructions executes in main

```
int main(){
    math xa(1),xb(2),xc(3);
    ++xa;
    xa.out();
    xa=xb+xc;
    xa.out();
    return 0;}
```

~~c~~ class math {

private: int a=0;

public:

math() { a=5; }

math(int b) { a=b; }

void operator++() { a+=2; }

void out() { cout << "The value of a is: " << this->a << endl; }

void operator+(math b, math c) { this->a = b->a + c->a; }

};